

HW1 - STAT 701

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Before Reading:

To make the proof more readable, I will miss or gap some natural or not important facts or notations during my writing. If you feel it hard to see, you can refer the appendix after the proof, where I will try to explain some simple conclusions (will be marked) more clearly. In case that you misunderstand the mark, I will add the mark just after those formulas between \$ and before those between \$\$.

And I have to claim that the appendix is of course a part of my assignment, so the reference of it is required. Enjoy your grading!

Problem 1. Ex. 1.3.

Suppose that $m_t = c_0 + c_1 t + c_2 t^2$, $t = 0, \pm 1, \dots$,

(a) Show that

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

where $a_2 = a_{-2} = -\frac{3}{35}$, $a_1 = a_{-1} = \frac{12}{35}$, $a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}$, $b_2 = b_{-2} = \frac{3}{21}$, $b_1 = b_{-1} = \frac{6}{21}$, $b_0 = \frac{7}{21}$.

(b) Suppose that $X_t = m_t + Z_t$, where $(Z_t)_{t \in \mathbb{Z}}$ is an independent sequence of normal random variables, each with mean 0 and variance σ^2 . Let

$$U_t = \sum_{i=-2}^2 a_i X_{t+i} \quad \text{and} \quad V_t = \sum_{i=-3}^3 b_i X_{t+i}$$

then

- (i) Find the means and variances of U_t and V_t .
- (ii) Find the correlations between U_t and U_{t+1} and between V_t and V_{t+1} .
- (iii) Which of the two filtered series $(U_t)_{t \in \mathbb{Z}}$ and $(V_t)_{t \in \mathbb{Z}}$ would you expect to be smoother in appearance?

Sol.

(a) Compare the coefficient of c_i for $i = 0, 1, 2$ respectively, we get

$$a_{-2} + a_{-1} + a_0 + a_1 + a_2 = 1,$$

$$(t-2)a_{-2} + (t-1)a_{-1} + ta_0 + (t+1)a_1 + (t+2)a_2 = t,$$

$$(t-2)^2 a_{-2} + (t-1)^2 a_{-1} + t^2 a_0 + (t+1)^2 a_1 + (t+2)^2 a_2 = t^2,$$

and

$$b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 = 1,$$

$$(t-3)b_{-3} + (t-2)b_{-2} + (t-1)b_{-1} + tb_0 + (t+1)b_1 + (t+2)b_2 + (t+3)b_3 = t$$

$$(t-3)^2b_{-3} + (t-2)^2b_{-2} + (t-1)^2b_{-1} + t^2b_0 + (t+1)^2b_1 + (t+2)^2b_2 + (t+3)^2b_3 = t^2$$

if and only if

$$m_t = \sum_{i=-2}^2 a_i m_{t+i} = \sum_{i=-3}^3 b_i m_{t+i}, \quad t = 0, \pm 1, \dots$$

Simplify the two systems above, we obtained

$$a_{-2} + a_{-1} + a_0 + a_1 + a_2 = 1,$$

$$-2a_{-2} + -a_{-1} + a_1 + 2a_2 = t,$$

$$4a_{-2} + 1a_{-1} + a_0 + 1a_1 + 2a_2 = t^2,$$

and

$$b_{-3} + b_{-2} + b_{-1} + b_0 + b_1 + b_2 + b_3 = 1,$$

$$-3b_{-3} + -2b_{-2} + -b_{-1} + b_1 + 2b_2 + 3b_3 = t,$$

$$9b_{-3} + 4b_{-2} + b_{-1} + b_1 + 4b_2 + 9b_3 = t^2$$

equivalently. Substituting $a_2 = a_{-2} = -\frac{3}{35}$, $a_1 = a_{-1} = \frac{12}{35}$, $a_0 = \frac{17}{35}$, and $b_3 = b_{-3} = -\frac{2}{21}$, $b_2 = b_{-2} = \frac{3}{21}$, $b_1 = b_{-1} = \frac{6}{21}$, $b_0 = \frac{7}{21}$ will verify the simplified systems above and we are done.

(b) (i) We know

$$\mathbb{E}U_t = \sum_{i=-2}^2 a_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t$$

$$\mathbb{E}V_t = \sum_{i=-3}^3 b_i \mathbb{E}(m_{t+i} + Z_{t+i}) = m_t$$

and

$$\text{Var}U_t = \mathbb{E} \left(\sum_{i=-2}^2 a_i (m_{t+i} + Z_{t+i}) - m_t \right)^2$$

$$= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right)^2 = \sum_{i=-2}^2 a_i^2 \sigma^2 = \sigma^2$$

$$\text{Var}V_t = \mathbb{E} \left(\sum_{i=-3}^3 b_i (m_{t+i} + Z_{t+i}) - m_t \right)^2$$

$$= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right)^2 = \sum_{i=-3}^3 b_i^2 \sigma^2 = \sigma^2$$

(ii) Notice that

$$U_t - \mathbb{E}U_t = \sum_{i=-2}^2 a_i Z_{t+i}, \quad V_t - \mathbb{E}V_t = \sum_{i=-3}^3 b_i Z_{t+i},$$

and hence

$$\begin{aligned}
\text{Cov}(U_t, U_{t-1}) &= \mathbb{E} \left(\sum_{i=-2}^2 a_i Z_{t+i} \right) \left(\sum_{i=-2}^2 a_i Z_{t+1+i} \right) \\
&= a_{-1}a_{-2} + a_0a_{-1} + a_1a_0 + a_2a_1 = \frac{48}{165} \\
\text{Cov}(V_t, V_{t-1}) &= \mathbb{E} \left(\sum_{i=-3}^3 b_i Z_{t+i} \right) \left(\sum_{i=-3}^3 b_i Z_{t+1+i} \right) \\
&= b_{-2}b_{-3} + b_{-1}b_{-2} + b_0b_{-1} + b_1b_0 + b_2b_1 + b_3b_2 = \frac{12}{49}
\end{aligned}$$

(iii) The covariance of V_t, V_{t+1} is smaller than that of U_t, U_{t+1} , and since V_t, U_t have equal variances, we may know that U_t, U_{t+1} are more dependent with each other than V_t, V_{t+1} , so $(U_t)_{t \in \mathbb{Z}}$ will be smoother.

Problem 2. Ex. 1.7.

Let $Z_t, t = 0, \pm 1, \dots$, be independent normal random variables, each with mean 0 and variance σ^2 , and let a, b , and c be constants.

Which, if any, of the following processes are stationary? For each stationary process, specify the mean and autocovariance function.

(a)

$$X_t = a + bZ_t + cZ_{t-1}.$$

(b)

$$X_t = a + bZ_0.$$

(c)

$$X_t = Z_1 \cos(ct) + Z_2 \sin(ct).$$

(d)

$$X_t = Z_0 \cos(ct).$$

(e)

$$X_t = Z_t \cos(ct) + Z_{t-1} \sin(ct).$$

(f)

$$X_t = Z_t Z_{t-1}.$$

Sol.

It is easy to check that for all the X_t above, $X_t \in L^1$ for any $t \in \mathbb{Z}$, we only check the existence of the autocovariance function for each process.

(a) Notice

$$\begin{aligned}
\gamma_X(t, t+h) &= \mathbb{E}(bZ_t + cZ_{t-1})(bZ_{t+h} + cZ_{t+h-1}) = (b^2 + c^2)\gamma_Z(h) + bc(\gamma_Z(h+1) + \gamma_Z(h-1)) \\
&= (b^2 + c^2)\delta_0(h) + bc(\delta_1 + \delta_{-1}(h)),
\end{aligned}$$

which is constant about t and hence X_t is a stationary process, with the autocovariance function above.

(b) Notice

$$\gamma_X(t, t+h) = \mathbb{E}b^2 Z_0^2 = b^2,$$

which is constant about t and hence X_t is a stationary process, with the autocovariance function above.

(c) Notice

$$\begin{aligned}\gamma_X(t, t+h) &= \mathbb{E}(Z_1 \cos(ct) + Z_2 \sin(ct))(Z_1 \cos(c(t+h)) + Z_2 \sin(c(t+h))) \\ &= \cos(ct) \cos(c(t+h)) + \sin(ct) \sin(c(t+h)) \\ &= \cos(ch),\end{aligned}$$

which is constant about t and hence X_t is a stationary process, with the autocovariance function above.

(d) Notice

$$\gamma_X(t, t) = \mathbb{E}Z_0^2 \cos^2(ct) = \cos^2(ct)$$

which is not constant about t and hence X_t is not a stationary process. (e) Notice

$$\gamma_X(t, t+1) = \mathbb{E}(Z_t \cos(ct) + Z_{t-1} \sin(ct))(Z_{t+1} \cos(ct+c) + Z_t \sin(ct+c)) = \cos(ct) \sin(c(t+1))$$

which is not constant about t and hence X_t is not a stationary process.

(f) Notice

$$\gamma_X(t, t+h) = \mathbb{E}Z_t Z_{t-1} Z_{t+h} Z_{t+h-1} = \delta_0(h)$$

which is constant about t and hence X_t is a stationary process, with the autocovariance function above.

Problem 3. Ex. 1.13.

Let $\{S_t : t = 0, 1, 2, \dots\}$ be the random walk with constant drift μ , defined by

$$S_0 = 0, \quad \text{and} \quad S_t = \mu + S_{t-1} + X_t, \quad t = 1, 2, \dots,$$

where $X_1, X_2, \dots$ are i.i.d. random variables with mean 0 and variance σ^2 . Compute the mean of S_t and the autocovariance function of the process $\{S_t\}$. Show that $\{\nabla S_t\}$ is stationary and compute its mean and autocovariance function.

Sol.

Notice

$$\mathbb{E}S_t = \mu + \mathbb{E}S_{t-1}, \quad S_0 = 0$$

and hence $\mathbb{E}S_t = t\mu$ for $t \in \mathbb{N}$. We know

$$S_t = t\mu + \sum_{i=1}^t X_i$$

as well, then we know

$$\gamma_S(t, s) = \mathbb{E}\left(\sum_{i=1}^t X_i\right)\left(\sum_{i=1}^s X_i\right) = (t \wedge s)\sigma^2.$$

We know

$$\nabla S_t = S_t - S_{t-1} = \mu + X_t =: Y_t, \quad \mathbb{E}Y_t = \mu$$

and it is easy to check $Y_t \in L^1$ and notice

$$\gamma_Y(t, t+h) = \delta_0(h)$$

which is constant about t and hence Y_t is a stationary process, with the autocovariance function above.

Problem 4. Ex. 1.17.

If

$$\mathbf{X} = (X_1, \dots, X_n)'$$

is a random vector with covariance matrix Σ , show that Σ is singular if and only if there exists a nonzero vector

$$\mathbf{b} = (b_1, \dots, b_n)' \in \mathbb{R}^n$$

such that

$$\text{Var}(\mathbf{b}'\mathbf{X}) = 0.$$

Sol.

For some $\mathbf{b} = (b_1, \dots, b_n)' \in \mathbb{R}^n$, we know

$$\text{Var}(\mathbf{b}'\mathbf{X}) = \mathbf{b}'\Sigma\mathbf{b}$$

and then we know if Σ is singular, then we may choose $\mathbf{b}$ nonzero such that $\mathbf{b}'\Sigma = \mathbf{0}$ and then $\text{Var}(\mathbf{b}'\mathbf{X}) = \mathbf{b}'\Sigma\mathbf{b} = 0$.

If there exists $\mathbf{b}$ nonzero such that $\text{Var}(\mathbf{b}'\mathbf{X}) = \mathbf{b}'\Sigma\mathbf{b} = 0$, assume $\Sigma = Q^T\Lambda Q$ a decomposition by the Spectral Theorem, where Λ is a diagonal matrix with nonnegative components by the nonnegative-definiteness of Σ , then if any diagonal component of Λ is positive, then we know

$$\mathbf{b}'\Sigma\mathbf{b} \geq a|Q\mathbf{b}|^2 > 0$$

where a is the minimum of the diagonal components of Λ which is a contradiction, and hence $Q^T e_i \in \ker \Sigma$ where the i th diagonal component of Λ is 0, which means Σ is singular.

Problem 5. Sample Autocovariances

Say I give you a time series $(X_1, \dots, X_n)$ that you know is stationary. Prove that the sample autocovariance function

$$\hat{\gamma}(h) = n^{-1} \sum_{t=1}^{n-|h|} (X_{t+|h|} - \bar{X})(X_t - \bar{X})$$

for $|h| < n$ gives a positive semi-definite sample autocovariance matrix

$$\hat{\Sigma}_{j,k} = \hat{\gamma}(|j-k|).$$

Sol.

Notice that define

$$Y_t = \begin{cases} X_t - \bar{X}, & 1 \leq t \leq n \\ 0, & \text{else} \end{cases}$$

and then

$$n\hat{\gamma}(|j-k|) = \sum_{t \in \mathbb{Z}} Y_{t+j} Y_{t+k}.$$

For $\mathbf{a} = (a_1, \dots, a_n)^T \in \mathbb{R}^n$, we know

$$\mathbf{a}^T \hat{\Sigma} \mathbf{a} = \mathbf{a}^T \left(\sum_{j=1}^n a_j \hat{\gamma}(|j-i|) \right)_{i=1}^n = \sum_{i=1}^n \sum_{j=1}^n a_i \hat{\gamma}(|i-j|) a_j,$$

and

$$\begin{aligned} n \sum_{i=1}^n \sum_{j=1}^n a_i \hat{\gamma}(|i-j|) a_j &= \sum_{i=1}^n \sum_{j=1}^n \sum_{t \in \mathbb{Z}} a_i a_j Y_{t+i} Y_{t+j} \\ &= \sum_{t \in \mathbb{Z}} \left(\sum_{i=1}^n a_i Y_{t+i} \right)^2 \geq 0 \end{aligned}$$

and hence $\mathbf{a}^T \hat{\Sigma} \mathbf{a} \geq 0$.

Problem 6. Causality

Consider the AR(1) process

$$X_t = 2X_{t-1} + Z_t,$$

where Z_t is some white noise with variance σ^2 . Define

$$Z_t^* = \frac{1}{4} Z_t - \frac{3}{4} \sum_{j=1}^{\infty} 2^{-j} Z_{t+j}.$$

1. Prove that Z_t^* is a white noise. What is its variance?
2. Prove that

$$X_t = \frac{1}{2} X_{t-1} + Z_t^*.$$

Sol.

1. It is easy to check that $\mathbb{E} Z_t^* = 0$ and

$$\begin{aligned} \gamma_{Z^*}(t, t+h) &= \mathbb{E} \left(\frac{1}{4} Z_t - \frac{3}{4} \sum_{j=1}^{\infty} 2^{-j} Z_{t+j} \right) \left(\frac{1}{4} Z_{t+h} - \frac{3}{4} \sum_{j=1}^{\infty} 2^{-j} Z_{t+h+j} \right) \\ &= \begin{cases} \sigma^2 \left(\frac{1}{16} + \frac{9}{16} \sum_{j=1}^{\infty} 2^{-2j} \right), & h = 0 \\ \sigma^2 \left(-3 \cdot 2^{-|h|} / 16 + \frac{9}{16} \sum_{j=1}^{\infty} 2^{-2j} 2^{-|h|} \right), & \text{else} \end{cases} \\ &= \begin{cases} \frac{\sigma^2}{4}, & h = 0 \\ 0, & \text{else} \end{cases}. \end{aligned}$$

Therefore, Z_t^* is a white noise with variance $\sigma^2/4$.

2. Since $Z_t = X_t - 2X_{t-1}$, we know

$$\begin{aligned} Z_t^* &= \frac{1}{4}(X_t - 2X_{t-1}) - \frac{3}{4} \sum_{j=1}^{\infty} 2^{-j}(X_{t+j} - 2X_{t+j-1}) \\ &= \frac{1}{4}(X_t - 2X_{t-1}) - \frac{3}{4} \left(\sum_{j=1}^{\infty} 2^{-j} X_{t+j} - \sum_{j=0}^{\infty} 2^{-j} X_{t+j} \right) \\ &= X_t - \frac{1}{2}X_{t-1}, \end{aligned}$$

and we are done.

Problem 7. Data exploration

Please download and take a look at the CET time series from Canvas.

1. Plot the data in some coherent way. This includes using proper timestamps and labeling the axes with the appropriate units. Does the data look like a stationary process? Why or why not? Perform some exploratory analysis to support your conclusion.
2. Select a segment containing

$$n = 365$$

contiguous observations, with no gaps or missing values.

Estimate the sample autocorrelation function (ACF) for increasingly large lags k , and plot the results. Discuss what happens as k increases and explain what you think is causing this behavior.

3. Design a linear mean function for the data using quadratic terms. For example, construct the design matrix using the following pseudocode:

```
t = 1:365
X = [tj^k for tj in t, k in 0:2]
```

Fit the linear model

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

using this design matrix.

Plot the residuals

$$\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Comment on what you observe. Do the residuals look like a stationary process? If not, describe the structures that suggest nonstationarity. Repeat the previous exploratory analyses to support your conclusion.

- 4*. Expand the analysis to

$$n = 3 \cdot 365$$

observations.

Construct a new design matrix $\mathbf{X}$ containing:

- a constant term;

- a linear trend;
- the seasonal component

$$\cos\left(\frac{2\pi t}{365}\right);$$

- the seasonal component

$$\sin\left(\frac{2\pi t}{365}\right).$$

Fit the least-squares mean function again and examine the residuals.

Normally, fitting a trigonometric function requires estimating both its frequency and its phase, which appears to be a nonlinear problem. Explain why the sine-and-cosine design-matrix approach correctly accounts for the phase using an ordinary linear least-squares model.

Sol.

1.

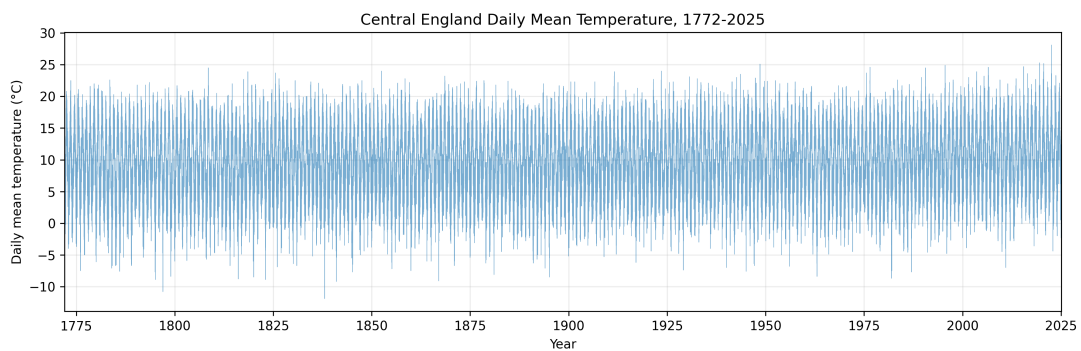


Figure 1: CET

Data points do not appear to be stationary. The average temperatures by month show a seasonal trend, with much lower temperatures in January than in July. They also differ across multi-decade periods, suggesting a changing mean level, which is contravertial to the invariant mean value of a stationary process.

2.

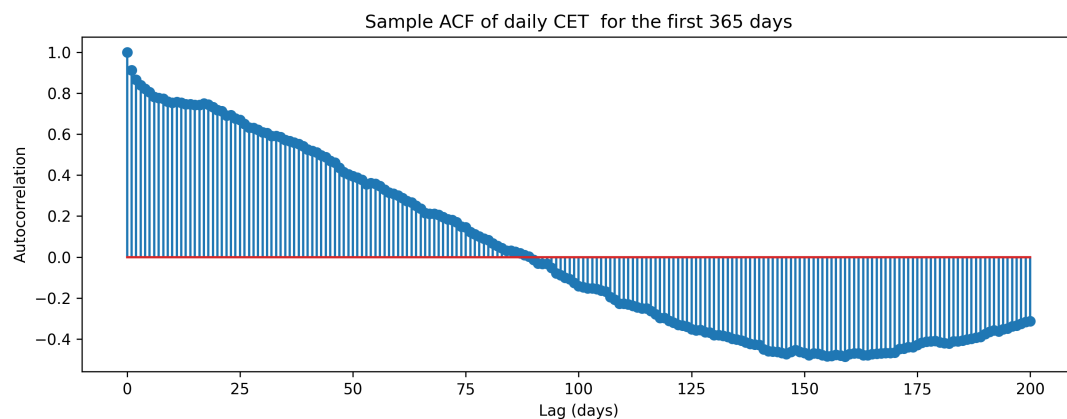


Figure 2: CET

As the lag k increases, the sample ACF decreases from positive values, crosses zero at approximately 90 days, and reaches a minimum of about -0.48 around 150-170 days before rising again. This pattern coincides an annual seasonal cycle: nearby days in the cycle tend to have similar temperatures, and observations roughly half a year apart often belong to opposite seasons and have deviations of opposite signs from the annual mean.

3.

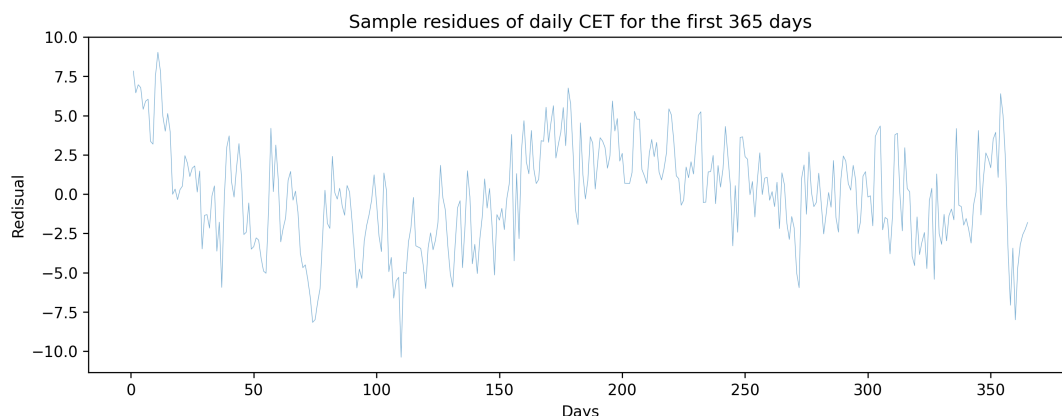


Figure 3: CET

After removing the fitted quadratic trend, the residuals fluctuate around zero, but they still exhibit sustained positive and negative runs. In particular, they are negative around days 40 – 150 and positive around days 160 – 240. This suggests that the quadratic trend may not fully describe the seasonal mean structure.

The means of four consecutive blocks are approximately -0.02 , -0.92 , 1.41 , and -0.47 , while their standard deviations range from 2.23 to 3.81. The residual ACF decays faster than that of the original series, but retains substantial short-lag correlation and becomes negative at longer lags.

Thus, the residuals appear closer to stationary than the original data, but remaining seasonal structure makes stationarity questionable. These results are suggestive rather than conclusive: autocorrelation alone does not imply nonstationarity, and one year of observations provides limited evidence. Examining a longer segment and explicitly modeling seasonality would help assess this further.

Problem 8. Trigonometric expansions

- a. Let Z_1 and Z_2 be i.i.d. $\mathcal{N}(0, 1)$ variables, and let $\omega \in [-\pi, \pi)$ be a constant. Define the process

$$Y_t = Z_1 \cos(\omega t) + Z_2 \sin(\omega t).$$

Calculate its ACF. Is it stationary? Is it strictly stationary?

- b. Pick n to be your favorite even integer, say 512 (although 1024 is also a very good even integer). Set

$$\omega_j = \frac{2\pi j}{n}, \quad j = 0, 1, \dots, n/2,$$

and simulate $(n/2 + 1)$ time series of length n given by

$$\begin{aligned} X_{0t} &= Z_0, & Z_0 &\sim \mathcal{N}(0, 1), \\ X_{jt} &= Z_{j1} \cos(\omega_j t) + Z_{j2} \sin(\omega_j t), & Z_{j1}, Z_{j2} &\sim \mathcal{N}(0, 2), \quad 1 \leq j < n/2, \\ X_{n/2,t} &= Z_{n/2}(-1)^t, & Z_{n/2} &\sim \mathcal{N}(0, 1). \end{aligned}$$

Now define a new process by

$$X_t = n^{-1/2} \sum_{j=0}^{n/2} X_{jt}.$$

1. Plot your data side-by-side with a white noise series of n i.i.d. $\mathcal{N}(0, 1)$ variables. What do you see?
2. Derive the ACF for X_t .
3. Play around with generalizing to

$$\text{Var}(Z_{j1}) = \text{Var}(Z_{j2}) = \sigma_j^2, \quad j \geq 1,$$

but otherwise not changing your code. Re-run everything and look at the resulting plot of your simulated process. Can you describe the effect of different choices for $\{\sigma_j^2\}$, for example, how things look if σ_j^2 increases or decreases with j ?

Sol.

a. We know

$$\begin{aligned} \gamma_Y(t, t+h) &= \mathbb{E}(Z_1 \cos(\omega t) + Z_2 \sin(\omega t))(Z_1 \cos(\omega(t+h)) + Z_2 \sin(\omega(t+h))) \\ &= \cos(\omega t) \cos(\omega(t+h)) + \sin(\omega t) \sin(\omega(t+h)) \\ &= \cos(\omega h), \end{aligned}$$

which is constant about t and hence Y_t is a stationary process, with autocorrelation function

$$\gamma_Y(h)/\gamma_Y(0) = \cos(\omega h).$$

For $\mathbf{t} := (t_1, \dots, t_k) \in \mathbb{Z}^k$, we know

$$\mathbf{Y}_{\mathbf{t}} := (Y_{t_1}, \dots, Y_{t_k})^T = \begin{bmatrix} \cos(\omega t_1) & \sin(\omega t_1) \\ \vdots & \vdots \\ \cos(\omega t_k) & \sin(\omega t_k) \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$$

and hence $\mathbf{Y}_{\mathbf{t}}$ is a normal random vector with covariance matrix

$$\begin{bmatrix} \cos(\omega t_1) & \sin(\omega t_1) \\ \vdots & \vdots \\ \cos(\omega t_k) & \sin(\omega t_k) \end{bmatrix} \begin{bmatrix} \cos(\omega t_1) & \sin(\omega t_1) \\ \vdots & \vdots \\ \cos(\omega t_k) & \sin(\omega t_k) \end{bmatrix}^T = [\cos(\omega(t_i - t_j))]_{i,j=1}^k$$

and hence

$$\mathbf{Y}_{\mathbf{t}} \stackrel{d}{=} \mathbf{Y}_{\mathbf{t}+h}$$

where $\mathbf{t} + h := (t_1 + h, \dots, t_k + h)$ for any $h \in \mathbb{Z}$ since it is easy to check they have the same covariance matrix. Therefore, Y_t is strictly stationary as well.

b. 1.

2. We may write X_t explicitly as

$$X_t = \frac{1}{\sqrt{n}} \left(Z_0 + \sum_{j=1}^{n/2-1} (Z_{j1} \cos(\omega_j t) + Z_{j2} \sin(\omega_j t)) + Z_{n/2} (-1)^t \right)$$

and then

$$\begin{aligned} \gamma_X(t, t+h) &= \frac{1}{n} (1 + 2 \sum_{j=1}^{n/2-1} \cos(\omega_j h) + 1) \\ &= \frac{2}{n} \sum_{j=0}^{n/2-1} \cos\left(\frac{2\pi j h}{n}\right) = \begin{cases} \frac{1 + (-1)^h}{n}, & h \neq 2kn \text{ for some } k \in \mathbb{Z} \\ 1, & h = 2kn \text{ for some } k \in \mathbb{Z} \end{cases} \end{aligned}$$